10/2/2020

Roll No.	
Name	
Class & Section	

APEEJAY COMMON ANNUAL EXAMINATION, 2019-20

PHYSICS (042)

Time Allowed : 3.00 Hrs. **Class – XI** **Maximum Marks : 70**

General Instructions :

1. *All Questions are compulsory. There are 37 questions in all.*
2. *This question paper has four sections : Section A, Section B, Section C and Section D*
3. *Section A contains 20 questions of one mark each, Section B contains 7 questions of two marks each, Section C contains 7 questions of three marks each and Section D contains 3 questions of five marks each.*
4. *There is no overall choice. However, internal choices have been provided in 2 questions of one mark, 2 questions of two marks, 2 questions of three marks and 3 questions of five marks. You have to attempt only one of the choices in such questions.*
5. *You may use the following values of physical constants wherever necessary.*

   $c=3 \times 10^{8}$ m/s

   $h=6.63 \times 10^{-34}$ Js

   $e=1.6 \times 10^{-19}$ C

   $m_e = 9.1 \times 10^{-31}$ kg

   $m_n = 1.675 \times 10^{-27}$ kg

   $m_p = 1.673 \times 10^{-27}$ kg

   Avogadro's number $= 6.023 \times 10^{23}$ *per gram mole*

   Boltzzmann constant$= 1.38 \times 10^{-23}$ JK

Section-A

Directions (Q.Nos 1 - 10)

Select the most appropriate option from those given below each question.

1. Doppler's effect holds good for 1

(a) Only sound waves.

(b) Only electromagnetic waves.

(c) Both (a) and (b).

(d) None of these.

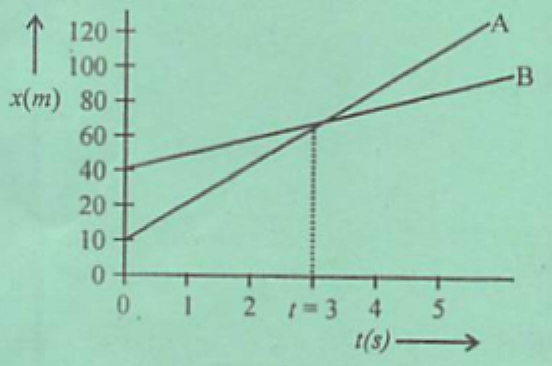
2. Two wave trains $Y_1 = \alpha \sin(4000\pi t)$ and $Y_2 = \alpha \sin(4008\pi t)$ are approaching each other. The number of beats heard per second is 1

(a) 8 (b) 4

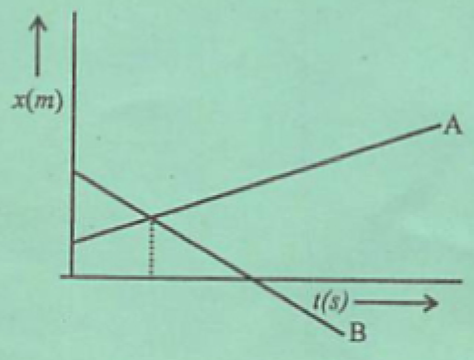
(c) 1 (d) 0

3. The average velocities of the objects A and B are V_A and V_B respectively. The velocities are related such that $V_A > V_B$. The position-time graph for this situation can be represented as 1

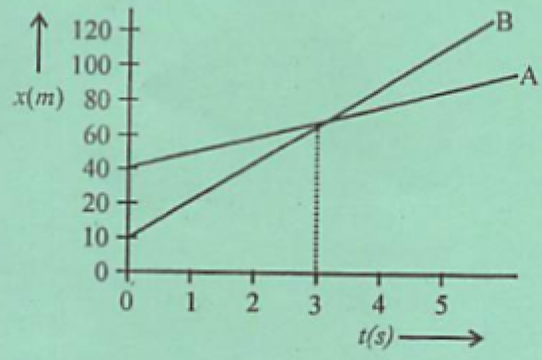
(a)



(b)



(c)

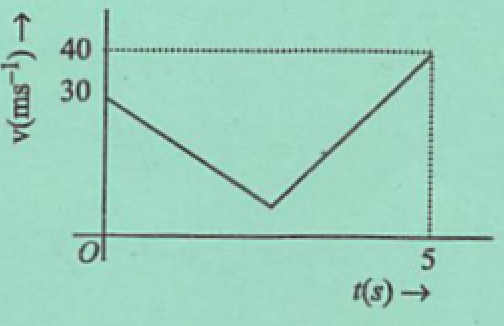


(d) None of these

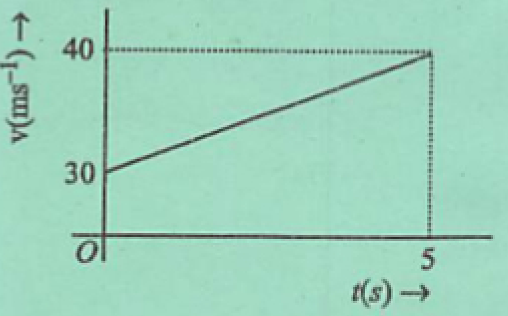
OR

An object is moving with an initial velocity of 30m/s with uniform acceleration. The velocity of the object increases to 40 m/s in next 5 s. The v-t graph which least represents this situation is

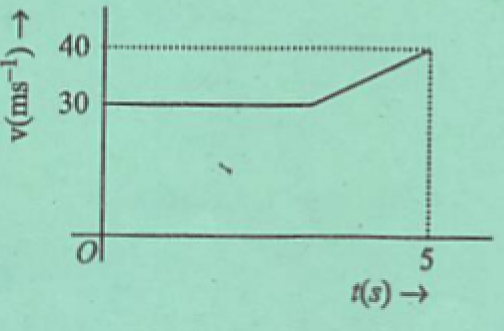
(a)



(b)



(c)



(d) None of these

4. Which of the following has unit but no dimension? 1

(a) Angle (b) Strain

(c) Relative velocity (d) Relative Density

5. A body falling freely under the action of gravity alone. Which of the following quantities remains constant during the fall?

(a) Kinetic energy (b) Potential energy

(c) Total mechanical energy (d) Total linear momentum

6. A wire is stretched to double its length. The strain is 1

(a) 2 (b) 1

(c) 0 (d) 0.5

7. Carnot engine is a 1

(a) irreversible engine (b) Petrol engine

(c) reversible engine (d) diesel engine

8. The PE i.e $U(x)$ can be assumed zero when (or at) 1

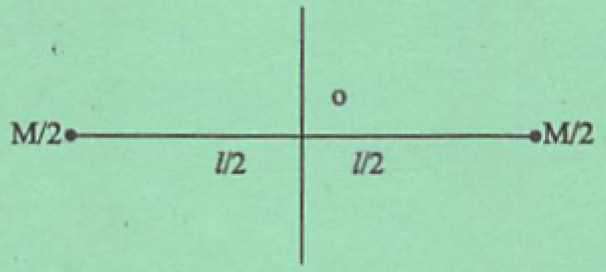
(a) $x = 0$.

(b) gravitational force is constant.

(c) infinite distance from the gravitational source.

(d) All of the above .

9. Two masses are joined with a light rod and the system is rotating about the fixed axis as shown in figure. 1



The MI of the system about the axis is

(a) $Ml^2/2$

(b) $Ml^2/4$

(c) Ml^2

(d) $Ml^2/6$

10. Change of state from solid to vapour state without passing through the liquid state is called : 1

(a) Regelation

(b) Sublimation

(c) Condensation

(d) Sedimentation

Directions (Q. Nos 11-15)

Answer in brief

11. Give the number of significant figures in 6.200×10^{10} sec. 1

12. The total momentum of universe remains constant. Is the statement true? State with reason. 1

In the following questions 13, 14 and 15, a statement is Assertion which is followed by a statement of Reason. Mark the correct choice as (a) Assertion is true statement, Reason is false statement or (b) Assertion and Reason both are true statements and Reason is the correct explanation Assertion or (c) Assertion and Reason both are true statements but Reason is not the correct explanation of Assertion or (d) Assertion is false statement but Reason is true statement or (e) Assertion and Reason both are false statements.

13. **Assertion :** A light body and a heavy body have same momentum. Then they also have same kinetic energy.

Reason : Kinetic energy does not depend on mass of the body. 1

14. **Assertion :** for a system of particles under central force field, the total angular momentum is conserved.

Reason : The torque acting on such system is zero. 1

15. **Assertion :** At the centre of earth, a body has centre of mass but no centre of gravity.
Reason : At the centre of earth, acceleration due to gravity $g = 0$. 1

Directions (Q.Nos 16-20)

Fill in the blanks :

16. The value of escape speed at the surface of a planet depends upon .... and 1

17. The angular momentum of mass m moving with velocity v along a circular path of radius r is given by 1

18. If two bodies stick together after a collision and move as a single body with a common velocity, then the collision is said to be 1

19. When the circular road is banked at an angle 0 and there is no friction between the road and the tyres of a car, $\mu = 0$, then the safe limit for maximum speed is 1

20. Two bodies are projected at an angle θ and $(\pi/2 - \theta)$ to the horizontal with the same speed. The ratio of their time of flight is 1

OR

A body is projected at an angle 45° with a velocity of 9.8 m/s. The horizontal range will be ....

Section-B

21. Find the dimensions of [a/b] in the equation $F = a\sqrt{x} + bt^2$, where F is force, x is distance and t is time. 2

22. Draw a velocity time graph for an object in uniform motion. Show that the area under the velocity time graph gives the displacement of the object in the given time interval. 2

OR

Two parallel rail tracks run north-south. Train A moves north with a speed of 54 km/h and train B moves south with a speed of 90 km/h. What is the relative velocity of B with respect to A?

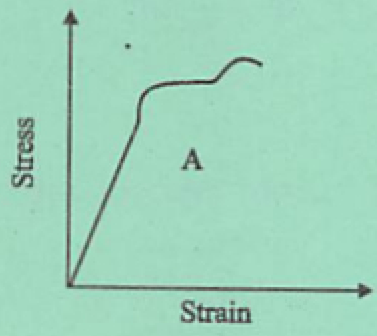
23. A block slides down on an inclined plane of inclination 30° with an acceleration equal to g/4. Find the coefficient of kinetic friction. **2**

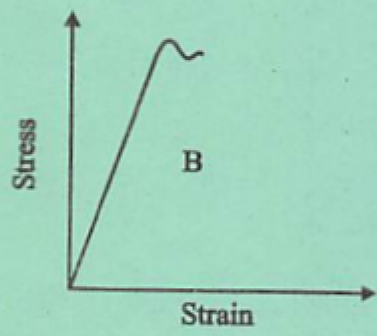
24. Show that the gravitational force is a conservative force. **2**

OR

Give important characteristics of elastic collision.

25. The stress strain graphs for materials A and B are shown in figure. **2**





The graphs are drawn on the same scale. State with reason.

1. Which of the material has greater Young's modulus?
2. Which of the two is the stronger material?

26. Two bodies of specific heats C_1 and C_2 having same heat capacities are combined to form a single composite body. What is the specific heat of the composite body? **2**

27. State Boyle's law on the basis of kinetic theory of gases. Draw P-V and P-1/V graph for a given mass of a gas at a constant temperature. **2**

Section-C

28. A Jet plane beginning its take off moves down the run way at a constant acceleration of 4 m/s^2. Find **3**

(i) The position and velocity of the plane 5 s after it begins to move.

(ii) If a speed of 70 m/s is required for the plane to leave the ground, how long the runway is required?

29. Derive Newton's third law of motion from the law of conservation of energy. **3**

OR

Define angle of friction. Deduce its relation with coefficient of friction.

30. Four bodies have been arranged at corners of a rectangle of length a and breadth b. Find the centre of mass of the system. 3

OR

State theorem of parallel axis with its expression. Find moment of inertia of a ring about a tangent in its plane.

31. A body weighs 63 N on the surface of the earth. What is the gravitational force on it due to the earth at a height equal to half the radius of the earth? 3

32. State Bernoulli's theorem. Write its expression. Apply Bernoulli principle to determine the speed of efflux from the side of a container when its top is open. 3

OR

What is capillarity? Derive an expression for the height to which the liquid rise in a capillary tube of radius r.

33. Explain why (or how)

(a) Bats can ascertain distances, directions, nature and sizes of the obstacles without any eyes?

(b) A violin note and sitar note may have the same frequency, yet we can distinguish between the two notes? 3

(c) Solids can support both longitudinal and transverse waves but only longitudinal waves can propagate in gases?

34. From a certain apparatus, the diffusion rate of hydrogen has an average value of $28.7 cm^3/s$. The diffusion of another gas under the same conditions is measured to have an average rate of $7.2 cm^3/s$. Identify the gas. 3

Section-D

35. Derive the expression for the kinetic and potential energies of a harmonic oscillator. Draw graphs for 5

(a) Energy vs. time

(b) Energy vs. displacement

OR

(a) What is the main difference between forced oscillation and resonance?

(b) What will be the change in time period of a loaded spring, when taken to moon?

(c) A spring of force constant 1200N/m is mounted horizontally on a horizontal table. A mass of 3 kg is attached to the free end of the spring, pulled sideways to a distance of 2 cm and released.

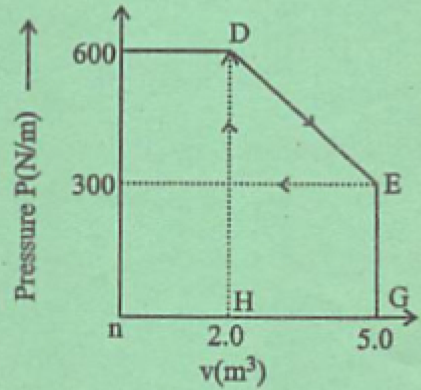
(i) What is the frequency of oscillation of the mass?

(ii) What is the maximum acceleration of the mass.

36. (a) Explain principle of heat engine. Find the expression of its efficiency.

(b) A thermodynamic system is taken from an original state to an intermediate state by the linear process as shown in the figure. Its volume is then reduced to the original value from E to F by an isobaric process. **5**

Calculate the total work done the gas from D to E to F.



OR

(a) State first law of thermodynamics with its expression.

(b) 500 J of work is done on a gas to reduce its volume by compression adiabaticall. What is the change in internal energy of the gas?

37. (a) Define centripetal acceleration. Derive an expression for centripetal acceleration of a particle moving with uniform speed v along a circular path of radius r.

(b) Find the magnitude of the centripetal acceleration of a particle on the tip of a fan blade 0.30 m in diameter, rotating at 1200 rev/minute.

OR

State parallelogram law of vector addition. Show that the resultant of two vectors A and B inclined at an angle θ is $R = \sqrt{A^2 + B^2 + 2AB \cos\theta}$. Discuss the special cases when (i) $\theta = 0°$ (ii) $\theta = 180°$ and (iii) $\theta = 90°$

ROLL NO.	
NAME	
CLASS & SECTION	

APEEJAY COMMON ANNUAL EXAMINATION, 2018-19 **015**

CLASS-XI

PHYSICS (042)

Time allowed : 3 hrs. *Maximum Marks : 70*

General Instructions :

(i) All questions are compulsory. There are 27 questions in all. This question paper has four sections: Section A, Section B, Section C and Section D.

(ii) Section A contains five questions of one mark each, Section B contains seven questions of two marks each, Section C contains twelve questions of three marks each, and Section D contains three questions of five marks each.

(iii) There is no overall choice. However, internal choices have been provided in two questions of one mark, two questions of two marks, four questions of three marks and three questions of five marks weightage. You have to attempt only one of the choices in such questions.

(iv) You may use the following values of physical constants wherever necessary.

$c = 3\times10^{8}$ m/s
$h = 6.63\times10^{-34}$ Js
$e = 1.6\times10^{-19}$ C
$\mu o = 4\Pi\times10^{-7}$ TmA^{-1}
$1/4\Pi\epsilon_0 = 9\times109$ Nm^{2} C^{-2}
$m_e = 9.1\times10^{-31}$ kg
mass of neutron $= 1.675\times10^{-27}$ kg
mass of proton $= 1.673\times10^{-27}$ kg
Avogadro's number $= 6.023\times10^{23}$ per gram mole
Boltzmann constant $= 1.38\times10^{-23}$ JK^{-1}

SECTION–A

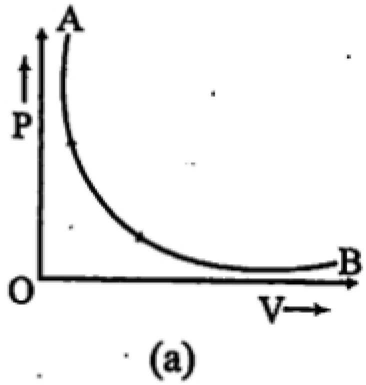
1. The Earth moving round the Sun in a circular orbit is acted upon by a force, and hence work must be done on the Earth by this force. Do you agree with this statement?

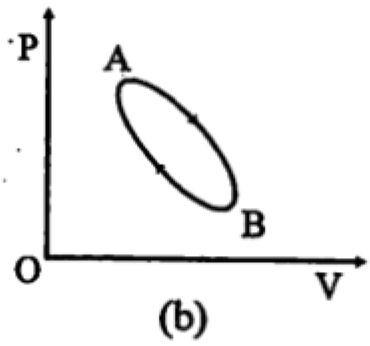
P.T.O.

OR

Give two examples from your daily life where according to Physics work done is zero. (1)

2. Which of the following processes shown here is reversible? Name the other process. (1)





OR

Write the expression of workdown in an (i) adiabatic process (ii) isothermal process.

3. An object is dropped from the top of a tower. Draw its velocity–time graph and displacement– time graph. (1)

4. Two lenses of same mass and same radius are given. One lens is convex and other is concave. Which one will have greater moment of inertia, when rotating about an axis perpendicular to the plane and passing through the centre? (1)

5. Prove that range is same for projection angles $(45 + \alpha)$ and $(45 - \alpha)$. (1)

SECTION–B

6. Does the function $y = \sin \omega t + \cos \omega t$ represent a periodic motion? What is the period of the motion?

OR

What is the beat frequency when two tuning forks of frequency 200 Hz and 205 Hz are sounded together? Mention one application of beats. (2)

7. The velocity time relation of an electron starting from rest is given by $v = kt$, where $k = 2\,\text{m/s}^2$. Calculate the distance travelled in 3 seconds. (2)

8. If the density ρ, acceleration due to gravity g and frequency υ are the basic quantities, find the dimensions of force in terms of ρ, g and υ.

OR

Distinguish between terms precision and accuracy of a measurement with the help of an example. (2)

9. What are conservative and non-conservative forces? Give one example of each. (2)

10. Explain if the ice in the polar caps of the earth melts how will it affect the duration of the day? (2)

11. Explain the principle behind the mirage formation in deserts, with the help of a ray diagram. (2)

12. The acceleration due to gravity on the surface of the moon is 1.7 m/s^2. What is the time period of a simple pendulum on the moon if its time period on the earth is 3.5 second? $g = 9.8 \text{m}/\text{s}^2$. (2)

SECTION–C

13. (a) What is Doppler effect?

(b) A whistle is being rotated in a horizontal circle. What will be the effect on the sound frequency for a listener standing (i) outside the circle (ii) at the centre of the circle. (3)

OR

Explain beats with the help of its graphical diagram ? Prove that beat frequency is given by the difference in frequencies of the two sound notes. (3)

14. A rain drop of radius 2 mm falls from a height of 500 m above the ground. It falls with decreasing acceleration (due to viscous resistance of the air) until at half its original height, it attains its maximum (terminal) speed, and moves with uniform speed thereafter. What is the work done by the gravitational force on the drop in the first and second half of its journey? What is the work done by the resistive force in the entire journey if its speed on reaching the ground is 10 m/s. (3)

15. A physical quantity X is given by : $X = \frac{P^2}{R^4}\frac{Q^{3/2}}{S^{1/2}}$. The percentage error in P, Q, R and S are 1%, 2%, 4% and 2%. Calculate the percentage error in X. (3)

16. Using velocity time graph, prove that $v^2 - u^2 = 2as$. Where symbols have their usual meanings. (3)

17. State parallelogram law of vector addition. Show that resultant of two vectors, vector A and vector B inclined at an angle θ is $R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$. (3)

18. What is the turning effect of force called ? On what factors does it depend ? Derive an expression for the centre of mass of a two particle system. (3)

OR

A meter stick is balanced on a knife edge at its centre. When two coins, each of mass 5g is put one on top of the other at the 12 cm mark, the stick is found to be balance at 45 cm. What is the mass of the meter stick? (3)

19. A sphere of metal A of 5 kg placed for sufficient time in a vessel containing boiling water, so that the sphere is at 100°C. It is then immediately transferred to 2 kg copper calorimeter containing 1 kg of water at 20°C. The temperature of water rises and attains a steady state at 50°C. Calculate the specific heat capacity of metal A. (Given : Specific heat of copper = 386 $Jkg^{-1}K^{-1}$, specific heat of water = 4186 $Jkg^{-1}K^{-1}$) (3)

OR

Explain what happens when the load on a metal wire suspended from a rigid support is gradually increased. Illustrate your answer with a suitable stress-strain graph.

20. State Bernoulli's principle for the flow of non-viscous fluids. Write its expression. Also explain how Bernoulli's principle helps in explaining the design of wing of aeroplane. (3)

OR

Define angle of contact and Capillarity. Derive an expression for the ascent of a liquid in a capillary tube.

21. A body cools from 80°C to 50°C in 5 minutes. Calculate the time it takes to cool from 60°C to 30°C, the temperature of the surrounding is 20°C. (3)

22. With the help of a block diagram, explain the working principle of a refrigerator and obtain an expression for its coefficient of performance. (3)

23. What are degrees of freedom? Using law of equipartition of energy, find the specific heat capacity of monoatomic and diatomic gas. (3)

24. A prism is made of glass of unknown refractive index. A parallel beam of light is incident on a face of the prism. The angle of minimum deviation is measured to be 40°. What is the refractive index of the material of the prism? The refracting angle of prism is 60°. If the prism is placed in water predict the new angle of minimum deviation of a parallel beam of light. (3)

SECTION–D

25. (a) What is escape velocity? Obtain an expression for the escape velocity on earth.

(b) What do you mean by rotational and translational equilibrium ? Calculate the moment of inertia of a disc of mass M and radius R about one of its diameters ? (5)

OR

State Kepler's laws of planetary motion. Find the potential energy of a system of four particles each of mass m placed at the vertices of a square of side 1. Also obtained the potential at the centre of the square.

26. Draw a ray diagram for the formation of image of a distant object by an astronomical telescope in normal adjustment position. Deduce the expression for its magnifying power. What are the advantages of reflecting type telescope over refracting type ? (5)

OR

Derive expression for the lens maker's formula where the symbols have their usual meanings. State the assumptions used and the convention of signs used.

27. (a) Write the laws of limiting friction. (5)

(b) Define angle of friction.

(c) Define angle of repose.

(d) What is the relation between angle of friction and angle of repose ?

OR

(a) What do you mean by banking of roads? Why it is done ?

(b) Derive an expression for velocity of a car on a banked circular road having coefficient of friction μ, hence write the expression for optimum velocity.

BEST OF LUCK!

APEEJAY SCHOOL, SAKET
FIRST TERM EXAMINATION (2018-19)
CLASS-XI
PHYSICS

Time: 3 Hours **Max. Marks: 70**

GENERAL INSTRUCTIONS

1. All questions are compulsory.
2. The question paper consists of 27 questions.
3. The question paper has ***four*** sections: Section ***A*** comprises ***5*** Questions (1 marks each), Section ***B*** comprises ***7*** Questions (2 marks each), Section ***C*** comprises ***12*** Questions (3 marks each) and Section ***D*** comprises ***3*** Questions (5 marks each).
4. Questions from Section A are to be answered in one word or one sentence as per requirement.
5. Calculator is not permitted. You may ask for logarithm table if required.

SECTION – A

1. Find the dimensional formula of coefficient of viscosity (η). **(1)**

2. Give an example of uniformly accelerated linear motion. **(1)**

3. What is the value of m in $\boldsymbol{i} + m\,\boldsymbol{j} + \boldsymbol{k}$ to be unit vector? **(1)**

4. What is the loss in kinetic energy after collision if the target of a body is initially at rest?**(1)**

5. Does the moment of inertia of a rigid body change with the speed of rotation? **(1)**

SECTION – B

6. Write the dimensions of (a) Moment of Inertia (b) Stress and Strain. **(2)**

OR

Using the relation , obtain the dimensions of Planck's constant .

7. Find the angle of projection at which horizontal range and maximum height are equal in case of a projectile. **(2)**

8. A woman throws an object of mass 500 g with a speed of 25 m/s.

i. What is the impulse imparted to the object?
ii. If the object hits a wall and rebounds with half the original speed, what is the change in momentum of the object? **(2)**

9. A spring balance reads forces in Newtons. The scale is 20 cm long and read from 0 to 60 N. Find the potential energy of spring when the scale reads 20 N. **(2)**

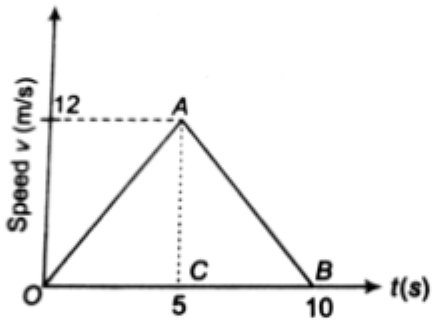
10. If the earth contracts to half of its radius. What would be the length of the day? **(2)**

11. In HCl molecule separation of nuclei of the two atoms is about 1.27 Å. Find the centre of mass of molecule given that chlorine atom is about 35.5 times massive than a hydrogen atom. **(2)**

12. Determine the speed with which the earth would have to rotate on its axis so that a person on the equator would weigh $3/5^{th}$ as much as his weight at present. Take the equatorial radius as 6400 km. **(2)**

SECTION – D

25. The speed-time graph of a particle moving along a fixed direction is shown in the figure below. Obtain the distance travelled by the particle between (i) t = 0 s to 10 s (ii) t =2 s to 6 s. What is the average speed of the particle over the intervals in (i) and (ii)? **(5)**



OR

Show that for motion of a car in a banked road the angle of banking θ for minimum wear and tear of tyres is given by where v is the velocity, r is the radius and g is acceleration due to gravity.

26. Show that the path of projectile is parabolic in nature using right expression. Derive an expression of horizontal range of projectile. **(5)**

OR

A particle of mass m moving with an initial velocity u collides inelastically with a particle of mass M initially at rest. If the collision is completely inelastic, find the expressions for

i. final velocity of the combined entity and
ii. loss in kinetic energy during collision.

27. Choose the correct alternatives :(Substantiate your answers with suitable explanation)

i. Acceleration due to gravity increase / decreases with increasing altitude. **(5)**
ii. Acceleration due to gravity increase / decreases with increasing depth(assuming earth to be uniform sphere)
iii. Acceleration due to gravity is independent of mass of the earth / mass of the body.
iv. The formula is more / less accurate than the formula)for the difference of potential energy between two points and distance away from centre of earth.

OR

Find the centre of mass for a solid cone of the base radius r and height h.

Questions Paper 2016-17
Session Ending Examination
CBSE Class XI Physics

General Instructions:

- All questions are compulsory.
- There is no overall choice. However, an internal choice has been provided in one question of two marks, one question of three marks and all three questions of five marks. You have to attempt only one of the choices in such questions.
- Questions number 1 to 5 are very short answer type questions, carrying one mark each.
- Questions number 6 to 10 are short answer type questions, carrying two marks each.
- Questions number 11 to 22 are also short answer type questions, carrying three marks each.
- Question number 23 is value-based question, carrying four marks.
- Questions number 24 to 26 are long answer type questions, carrying five marks each.
- Use of calculators is not permitted. However, you may use log tables, if necessary: y G = 6.67×10^{-11} Nm^2/kg^2, Mass of earth M = 6×10^{24} kg, Radius 4 of earth R = 6.4×10^6 m, Acceleration due to gravity of earth g = 9.8 m/s^2

1. Write four fundamental forces in the ascending order of their strength. (1)

2. State Work Energy theorem. (1)

3. What are the factors on which center of mass of a body depends? (1)

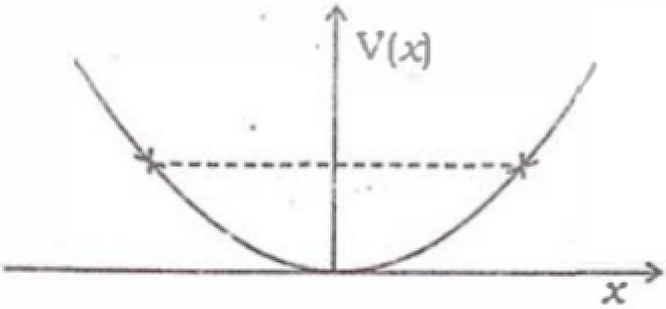
4. Define absolute zero temperature. (1)

5. What is the time period of second's pendulum? (1)

6. If E, M, J and G respectively denote energy, mass, angular momentum and gravitational constant, calculate the dimensions of EJ^2/M^5G^2. (2)

7. The potential energy function for a particle executing linear simple harmonic motion is

given by $V(x)$ = [illegible] , where k is the force constant of the oscillator. For $k = 0.5\ Nm^{-1}$, the graph of $V(x)$ versus x is shown in given fig. Show that a particle of total energy 1 J moving under this potential must turn back when it reaches = ±2 m. (2)



OR

The bob of a pendulum is released from a horizontal position. If the length of the pendulum is 1.5 m, what is the speed with which the bob arrives at the lowermost point, given that it dissipated 5% of its initial energy against air resistance?

8. Derive an expression for orbital velocity of a satellite. Establish a relation for orbital velocity of a satellite orbiting very close to the surface of earth. (2)

9. A pan filled with hot food cools from 94°C to 86°C in 2 minutes when the room temperature is at 20°C. How long will it take to cool from 71°C to 69°C? (2)

10. A Carnot engine whose heat sink is at 27° has an efficiency of 40%. By how many degrees should the temperature of source be changed to increase the efficiency by 10% of the original efficiency? (2)

11. A ball is thrown vertically upwards with a velocity of 20 m/ s from the top of a multistorey building. The height of the point from where the ball is thrown is 25.0 m from the ground. (3)

(a) How high will the ball rise?

(b) How long will it be before the ball hits the ground?

12. Define centripetal acceleration and find out an expression for centripetal acceleration. (3)

Question Paper 2017 set 1

CBSE Class 11 Physics

1. Define the term elasticity

2. Write the SI unit of Gravitational constant.

3. Fill up 3.0 m/s^2 =______km/hr^2

4. Under what condition is the relation s=ut is correct.

5. State relation between impulse and momentum

6. State Hooke's Law.

7. State Boyle's Law

8. Action and reaction forces do not balance each other why?

9. Calculate the degrees of freedom for monoatomic, diatomic and triatomic gas

10. What do you mean by mean free path and write its formula.

11. Derive an expression for the work done during Isothermal expansion

12. State the second law of thermodynamics and write its two applications of it.

13. The position of a particle is given by r=3.0 t $\hat{i} - 2.0t^2\hat{j} + 4.0\hat{k}$ unit metre where t in seconds and cofficients have the proper units for r to be in metres. Find the velocity and acceleration of the particle.

14. Two billiard balls each of mass 0.05kg moving in opposite direction with speed 6 m/s collied and rebound with the same speed. What is the impulse imparted to each ball due to other?

15. E, m, L and denote energy, mass, angular momentum and gravitational constant respectively. Determine the dimension of EL^2/m^5G^2

16. Define modulas of elasticity and write down its units and Dimensions.

17. Write down the relation between three types of moduli and poissons ratio.

18. The force acting on an object of mass m travelling of velocity v in a circule of radius r is giving by $f = mv^2/r$. The measurements recorded as m = 3.5kg ± 0.1 kg v = 20m/s ± 1m/s & r = 12.5m ± 0.5m find the maximum possible (i) fractional error (ii) percentage error in the measurement of force.

19. Define v = u + at from velocity--time graph.

20. How is centripetal force provides in case of the following?

(i) motion of planet around the sun

(ii) motion of moon around the sun

(iii) motion of an electron around the nucleus in an atom.

21. Determine the volume of 1 mole of any gas at S. T. P., assuming it behaves like an ideal Gas.

22. A carnot engine develops 100H.P. and operates between 27°C and 227°C.Find.

(i) Thermal effiency.

(ii) Heat supplied.

(iii) Heat rejected?

23. Explain and derive parallelogram law of vector additor.

24. State and prove the function of refrigerator. With the help pf labelled diagram.

25. A stone is projected at an angle ∅ with an initial velocity u under the effect of gravity then derive the expression for

(a) Horizontal range.

(b) Maximum height.

(c) Time of flight

26. (a) How does carnot cycle operates

(b) Why does absolute zero not correspond to zero energy

CBSE Class – XI

Physics (Set 1)

Last year Paper (2015-16)

Time: 3 Hrs. M.M: 70

General Instructions:

(i) Question 1 to 5 one mark.

(ii) Question 6 to 10 each two mark.

(iii) Question 11 to 22 each three mark.

(iv) Question 23 is value based question and carry four marks.

(v) Questions 24 to 26 each five mark.

Section A

1. Write the dimensional formula of impulse and name of the physical quantity having same dimension.

2. Write the relation between two angles for which horizontal ranges will be equal.

3. On what factors does the two vectors a and b to be perpendicular to each other.

4. Write the condition for conservation of mechanical energy of a system.

5. What is the ratio of escape speed of the earth and escape speed of the moon?

Section B

6. Define wavelength. Why there is no transfer of energy by standing waves?

7. Is the heat supplied to a system always equal to the increase in its internal energy?

8. Resultant of two vectors of equal amplitude is equal in magnitude with one of the vectors, calculate the angle between them

Or

A woman throws an object of mass 500 g with a speed of 25 m/s.

(i) What is the impulse imparted to the object?

(ii) If the object hits a wall and rebounds with half the original speed, what is the change in momentum of the object?

9. If length of a pendulum is increased by x%. Find error in time per day.

10. Define coefficient of linear expansion and coefficient of volume expansion of solid. How are they related?

Section C

11. The density of a linear rod of length L varies as $\rho = \alpha + \beta x$, where x is the distance from the left end. Locate the centre of mass.

12. Two particles each of mass m go round a circle of radius a under the action of their mutual gravitational attraction. Find the speed of each particle. Also analyse in terms of kinetic energy and gravitational potential energy.

13. Find the angle made by a cyclist with the vertical while taking a circular turn of radius a. Also state the forces which are providing centripetal acceleration and maintain vertical equilibrium.

14. What is the moment of inertia of a solid sphere of radius R about one of its diameter. If moment of inertia about the parallel tangent is I_0. Also, state the theorem of perpendicular axes.

15. Find the expression for potential energy of a strained body.

16. State Torricelli's law for efflux speed of a liquid and prove it. Find the efflux speed if height is 5 m.

17. Establish ideal gas equation in terms of gas constant R, volume V, pressure p, temperature T and number of moles n.

Or

For an oscillating pendulum, find the time period of oscillation.

18. Two separate air bubbles formed of the same liquid (surface tension T) come together to form a double bubble. Find the radius and the sense of curvature of the internal film surface common to both the bubbles.

19. The position of a particle is given by r= $\left(3t\hat{i} - 2t^2\hat{j} + 4\hat{k}\right) m$ where, t is in second and the coefficients have the proper units for r to be in metres.

(i) Find the v and a of the particle?

(ii) What is the magnitude and direction of velocity of the particle at t = 2 s?

20. A10 kW drilling machine is used to drill a bore in a small aluminium block of mass 8 kg.

How much is the rise in temperature of the block in 2.5 min, assuming 50% of the power is used up in heating the machine itself or lost to the surroundings. Specific heat of aluminium is 0.91 J/g-K.

21. Ten one-rupee coins are put on top of each other on a table. Each coin has mass m. Give the magnitude and direction of

(i) the force on the 7th coin (counted from the bottom) due to all the coins on its top,

(ii) the force on the 7th coin by the eighth coin,

(iii) the reaction of the 6th coin on the 7th coin.

22. A gas mixture consists of molecules of types A, B and C with masses $m_A > m_B > m_C$. Rank the three types of molecules in decreasing order of (i) average KE (ii) rms speeds.

Section D

23. John found his mother suffering from viral fever and took her to the doctor for treatment. While examining her, the doctor used a thermometer to know the temperature of the body. He kept the thermometer in the mouth of the patient and recorded the reading as $102°F$. Doctor gave the necessary medicines. After coming home, John asked his mother, why thermometer is used to know the temperature of body? Why mercury is used in a thermometer when there are so many liquids? Then, his mother explained him in detail.

(i) Comment upon the values displayed by John's mother.

(ii) A newly designed thermometer has its lower fixed point marked at $5°$ and $95°$, respectively. Compute the temperature on this scale corresponding to $50°$ C.

Section E

24. Two identical springs each of force constant k are connected in (i) series and (ii) parallel, so that they support a mass m. Find the ratio of the time periods of the mass in the two systems.

Or

Draw the first three harmonics in an open organ pipe. Two piano strings A and B are playing slightly out of tune and produce beats of frequency 5 Hz. The tension in string B is slightly increased and the beat frequency is found to decrease to 3 Hz. What is the original frequency of B if the frequency of A is 500 Hz?

CBSE Class – XI

Physics (Set 2)

Last year Paper (2015-16)

Time: 3 Hrs. M.M: 70

General Instructions:

(i) Question 1 to 5 one mark.

(ii) Question 6 to 10 each two mark.

(iii) Question 11 to 22 each three mark.

(iv) Question 23 is value based question and carry four marks.

(v) Questions 24 to 26 each five mark.

Section A

1. Identify the term which has unit but no dimension.

2. Would you expect a typical trajectory of a ping-pong ball to be a parabola, why?

3. What property of a tennis determine the ease with which it can be rapidly rotated?

4. Define safety factor in elasticity

5. Can water be made to boil without heating?

Section B

6. Two principal thrusts in physics are unification and reduction. Justify by giving illustrations.

7. An aircraft executes a horizontal loop at a speed of 720 km/h with its wings banked at $15°$. What is the radius of the loop?

8. Can a body have energy without having momentum and have momentum without energy? Explain.

9. Pressure of a gas in a closed cylinder is expressed in the following manner p = $p_a + h\rho g$

Identify the expressions for

(i) absolute pressure of the gas

(ii) gauge pressure of the gas

10. What do you mean by phase of a wave? How does it change with position?

Or

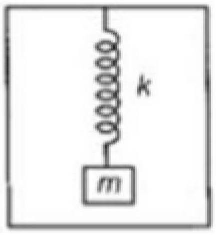
The frequencies of two tuning forks A and B are 250 Hz and 255 Hz, respectively. Both are sounded together. How many beats will be heard in 5s?

Section C

11. A particle of mass m is projected with velocity v at an angle θ with the horizontal. Find its angular momentum about the point of projection when it is at the highest point of its trajectory.

12. Find the expression for variation of acceleration due to gravity g, with height above the surface of the earth.

13. A spring mass system is hanging from the ceiling of an elevator in equilibrium. The elevator suddenly starts accelerating upwards with acceleration a.



Find:

(i) the frequency

(ii) the amplitude of the resulting SHM.

14. What is an isothermal process? Derive an expression for work done during an isothermal expansion of a gas.

15. What is capillarity? Water rises in a capillary tube to a height of 4.0 cm. In another capillary tube whose radius is one third of it, how much the water will rise?

16. Compute the fractional change in volume of a glass slab, when subjected to a hydraulic pressure of 10 atm.

17. Find the molar specific heat of the process p = b for a diatomic gas, b being constant.

18. Define centre of mass of a n particles system. State the coordinates of centre of mass of a n particles system.

19. Derive the expression for excess pressure inside a spherical liquid drop of radius R.

Or

The molecules of a given mass of a gas have root mean square speeds of 100 m/s at $27°C$ and 1.00 atm pressure. What will be the root mean square speeds of the molecules of the gas at $127°C$ and 2.0 atm pressure?

20. State and establish the law of equipartition of energy.

21. A body oscillates with SHM according to the equation

$$x(t) = 10 \cos\left(2\pi t + \frac{\pi}{6}\right)$$

where x is in metre and t is in second.

Calculate

(i) displacement at time t = 0

(ii) magnitude of maximum velocity

(iii) angular frequency.

22. Between p-V diagrams corresponding to adiabatic and isothermal process which is steeper and why?

Section D

23. Sonu and Raju were playing cricket. Raju hit the ball bowled by Sonu. Sonu starting running with the ball in air. When ball comes near Sonu's hand, he caught the ball suddenly due to this his finger got injured. Raju took Sonu to hospital, where he was treated. Later, Raju explained Sonu why his finger got injured while he was catching the ball.

(i) What are values shown by Raju?

(ii) Why Sonu got injured?

(iii) What values we should learn from this incidence?

Section E

24. **(i)** Derive an expression for the acceleration of a body sliding down a rough inclined plane.

(ii) A block of mass M is pulled along a horizontal frictionless surface by a rope of mass m by applying a force F at the free end of the rope. Find the force by the rope on the block.

Or

A satellite is to be placed in equatorial geostationary orbit around the earth for communication.

(i) Calculate height of such a satellite.

(ii) Find out the minimum number of satellites that are needed to cover entire earth so that

CBSE Class – XI

Physics (Set 3)

Last year Paper (2015-16)

Time: 3 Hrs. M.M: 70

General Instructions:

(i) Question 1 to 5 one mark.

(ii) Question 6 to 10 each two mark.

(iii) Question 11 to 22 each three mark.

(iv) Question 23 is value based question and carry four marks.

(v) Questions 24 to 26 each five mark.

Section A

1. Find the dimensional formula for coefficient of viscosity (η).

2. Write properties of an ideal fluid.

3. If pressure is made 16 times, what will be the effect on the velocity of sound?

4. Given that for two vectors A and B, $|A\times B| = |A.B|$. Find the acute angle between A and B.

5. A body of mass m is oscillating harmonically suspended from a mass less spring of spring constant k. What is the time period?

Section B

6. If a = $2\hat{i}+3\hat{j}-4\hat{k}$ and b = $4\hat{i}+3\hat{j}-2\hat{k}$. Find the angle between a and b.

Or

Establish the given vector inequality geometrically or otherwise $|a+b|<|a|+|b|$ When does the equality sign above apply?

From the properties of a triangle, one side of a triangle is always less than the sum of the lengths of its two other sides.

7. Define inertial and non-inertial frame of references. What is pseudo force?

8. Solve the expression for potential energy of a spring when elongation in the spring is x.

9. Write expression for work done in blowing a soap bubble from radius r_1 to r_2. Calculate it to increase radius from r to 2r.

10. A Carnot's engine has the same efficiency

(i) between 500 K and 100 K

(ii) between 1000 K and T K

Find the value of T.

Section C

11. (i) Using the relation E = hv, obtain the dimensions of Planck constant.

(ii) The resistance R is given by relation

R = V/ I. If potential difference V is $100 \pm 5V$ and current I is 10 ± 0.2 A. Calculate the percentage error in R.

12. A car accelerates from rest to a constant rate a for some time, after which it decelerates at a constant rate p to come to rest. If t is the total time elapsed, then calculate

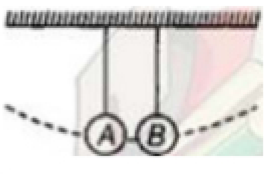
(i) the maximum velocity attained by the car.

(ii) the total distance travelled by the car.

13. A projectile is fired horizontally from the top of a tower. Find the expression for its time of descent and horizontal range.

14. Distinguish between static friction, limiting friction and kinetic friction. How do they vary with the applied force, explain by diagram.

15. Two pendulums with identical bobs and lengths are suspended from a common support such that in rest position the two bobs are in contact as shown in figure alongside. One of the bobs is released after being displaced by 10° so that it collides elastically head-on with the other bob.



(i) Describe the motion of two bobs.

(ii) Draw a graph showing variation in energy of either pendulum with time, for $0 \leq t \leq 2T$, where T is the time period of each pendulum.

16. (n-1) equal point masses each of mass m are placed at the vertices of a regular n polygon. The vacant vertex has a position vector a with respect to the centre of the polygon. Find the position vector of centre of mass.

CBSE Class – XI

Physics (Set 4)

Last year Paper (2015-16)

Time: 3 Hrs. M.M: 70

General Instructions:

(i) Question 1 to 5 one mark.

(ii) Question 6 to 10 each two mark.

(iii) Question 11 to 22 each three mark.

(iv) Question 23 is value based question and carry four marks.

(v) Questions 24 to 26 each five mark.

Section A

1. Find the dimensions of pressure.

2. A force of 10 N acts on a particle along a direction making an angle of $37°$ with the vertical. Find the component of the vertical direction.

3. A person travelling on a straight line with uniform velocity v_1 for some time and v_2 for next equal time. Find average velocity.

4. Mention the physical quantities, remain conserved in an inelastic collision?

5. Two spherical balls of mass M and radius R placed at distance a apart. Write expression for the force between them.

Section B

6. What is the heat associated with adiabatic process and what is the change in internal energy for isothermal process?

7. A ball is thrown vertically upwards. Draw its

(i) velocity-time curve.

(ii) acceleration-time curve.

8. Estimate will be angle of projection of a projectile for which range R and maximum height H are equal.

9. Obtain the equation, $\omega = \omega_0 \alpha t$.

Or

Why is the weight of a body at the poles more than the weight at the equator? Explain.

10. What is Kepler's law of periods? Express it mathematically.

Section C

11. (i) In van der Wiaal's equation

$$\left[p+\left(\frac{a}{V^2}\right)\right](V-b)=RT$$

what are the dimensions of a and b?

where p is pressure, V is volume, T is temperature and R is gas constant.

(ii) Find the relative error in Z, if $Z = \frac{A^4 B^{1/3}}{CD^{3/2}}$.

12. Two parallel rail tracks run North-South. Train A moves North with a speed of 54 km/h and train B moves South with a speed of 90 km/h. What is the

(i) velocity of B w.r.t. A?

(ii) velocity of ground with respect to B?

(iii) velocity of a monkey running on the roof of the train A against its motion (with a velocity of 18 km/h w.r.t. the train A) as observed by a man standing on the ground?

13. A projectile is fired with speed u making an angle 0 with horizontally from the surface of the earth. Prove that the projectile will hit the surface of the earth with same speed and at the same angle.

14. A block of mass m is held against a rough vertical wall by pressing it with a finger. If the coefficient of friction between the block and the wall is p, and the acceleration due to gravity is g, calculate the minimum force required to be applied by the finger to hold the block against the wall?

15. A railway car of mass 20 tonne moves with an initial speed of 54 km/h. On applying brakes, a constant negative acceleration of 0.3 m/s2 is produced.

(i) What is the braking force action on the railway car?

(ii) In what time will it stop?

(iii) What distance will be covered by railway car before it finally stops?

16. Establish a relation between angular momentum and moment of inertia of a rigid body. Define moment of inertia in terms of it.

17. Find an expression for the orbital speed of a satellite revolving around the earth in a circular orbit at a height h above the surface of the earth.

18. Draw stress-strain graph for a metallic solid and show

(i) elastic limit

(ii) region of plasticity

(iii) point of ultimate tensile strength

(iv) fracture point on the graph

Or

The sap in trees, which consists mainly of water in summer, rises in a system of capillaries of radius r = $2.5\times10^{-5}\,m$. The surface tension of sap is T = $7.28\times10^{-2}\,N/m$ and the angle of contact is $0°$. Does the surface tension alone account for the supply of water to the top of all trees?

19. Do State law of equipartition of energy. Use this law to calculate specific heats of monoatomic, diatomic and triatomic gases.

20. An insulated container containing monoatomic gas of molar mass m is moving with a velocity v_0. If the container is suddenly stopped, find the change in temperature.

21. Show that the motion of a particle represented by y = $\sin\omega t - \cos\omega t$ is simple harmonic with a period of $\frac{2\pi}{\omega}$.

22. What correction was applied by Laplace in Newton's formula for speed of sound waves? Does it lead to correct value of speed of sound in air?

Section D

23. We daily observe that motion of any object falling freely is continuously accelerated and the value of acceleration is found to be about 9.8 m/s2 near the earth's surface. However, rain drops falling due to condensation of water vapour in a cloud do not faR on the ground with an excessively high speed although on an average a cloud is moving at a height of more than-a kilometre from the earth's surface. Arvind's father did not know the cause of this. He thought that size and mass of rain drops are extremely small, negligible force of gravity acts on rain drop and so the falling speed of rain drop is quite small.

One day he asked his son Arvind about his thinking. Arvind explained the real cause that

CBSE Class – XI

Physics

Last year Paper (2015-16)

Time: 3 Hrs. M.M: 70

General Instructions:

(i) Question 1 to 5 one mark.

(ii) Question 6 to 10 each two mark.

(iii) Question 11 to 22 each three mark.

(iv) Question 23 is value based question and carry four marks.

(v) Questions 24 to 26 each five mark.

Section A

1. All constants are dimensionless. Comment.

2. When can an object in motion be considered as a point object?

3. A particle is moving in a straight line. Is it possible for it to maintain the motion in the same direction while the acceleration is in the reverse direction.

4. On what factors does the value of angle of contact depend?

5. Plot a graph showing the variation of specific heat of water with temperature.

Section B

6. What are the important steps involved in a scientific method to understand a natural phenomenon?

7. Determine the maximum acceleration of the train in which a box lying on its floor will remain stationary, given that the coefficient of the static friction between the box and the train's floor is 0.15.

8. State and explain work-energy theorem.

9. The excess pressure inside a soap bubble is thrice the excess pressure inside a moving soap bubble. What is the ratio between the volume of the first and second bubble?

10. State the principle of superposition of waves. What happens when phase differences between two waves are (i) 0 and (ii) π ?

Or

A harmonic oscillator is oscillating with a frequency of 3 Hz. Its acceleration amplitude is $0.36\pi^2 m/s^2$. Determine its velocity amplitude and the amplitude of displacement.

Section C

11. What are systematic errors? What is their cause? How can these be minimised?

12. A 100 m sprinter uniformly increases his speed from rest at the rate of $1m/s^2$ up to $\frac{3}{4}th$ of the total run and then covers the balance $\frac{1}{4}th$ run with uniform speed. 4 How much time does he take to complete the race?

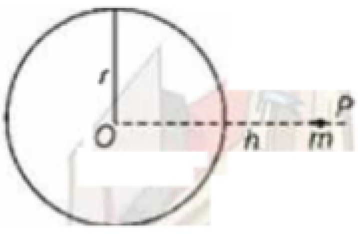
13. An object of mass m is moving in a circular motion of radius r at a constant speed v. Obtain an expression for the magnitude of acceleration of the object.

14. Find the instantaneous acceleration of a 1.0 kg mass suspended from a spring constant of force constant 5.0 N/cm, when the spring is stretched 10.0 cm. The mass is initially at rest.

15. How can you show that when a particle suffers an oblique elastic collision with another particle of equal mass and initially at rest, the two particles would move in mutually perpendicular directions after collision?

16. How is angular velocity related to linear velocity of a particle of the body in rotational motion? Does pure rotation take place at uniform velocity or at uniform linear velocity?

17. A mass m is placed at P a distance h along the normal through the centre O of a thin circular ring of mass m and radius r as shown in figure.



If the mass is removed further away such that OP becomes 2h, by what factor the force of gravitation will decrease, if h = r?

18. Two identical solid ball, one of irony and the other of wet clay are dropped from the same height on the floor. Which one will rise to a greater height after striking the floor and why?

Or

During summer in India, one of the common practice to keep cool is to make ice balls of crushed ice, dip it in flavoured sugar syrup and sip it. For this, a stick is inserted into crushed ice and is squeezed in the palm to make it into the ball. Equivalently in winter, in those areas where it shows, people make snow balls and throw around. Explain the formation of ball out of crushed ice or snow in the light of p-T diagram of water.

19. If the coefficient of performance of a refrigerator is 5 and operates at the room temperature $27°C$. Find the temperature inside the refrigerator.

20. Using equipartition law of energy, obtain the value of molar specific heat capacity of water. Does it agree with the experimental result?

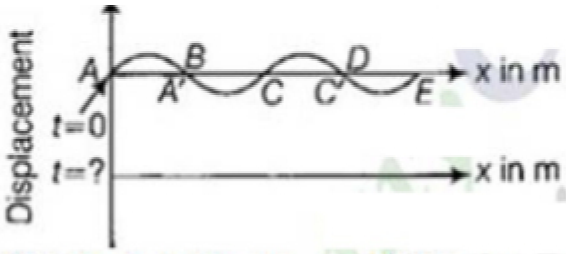
21. A particle is executing SHM with amplitude A and has a maximum velocity v_0.

(i) At what displacement will its velocity be $\frac{v_0}{2}$?

(ii) What is the velocity at displacement $\frac{A}{2}$?

22. The pattern of standing waves formed on a stretched string at two instants of time are shown below.

The velocity of two waves superimposing to form stationary waves is 360 m/s and their frequencies are 256 Hz.



(i) Calculate the time at which the second curve is plotted.

(ii) Make nodes and anti-nodes on the curve.

(iii) Calculate the distance between A' and C'.

Section D

23. Shikhaj went to a hill station with his family. On his way, he was fascinated by the beauty of mountains all around him. But a question started creeping in his mind that how such roads have been made in the mountains which seem so unapproachable. He asked his father the same question who told him about remote sensing satellites which are used to get information of places which can't be reached directly.

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CUMULATIVE EXAMINATION - 2014-15

SUBJECT - XI

CLASS - PHYSICS

Time : 3 Hrs. **M.M. 70**

General Instructions:

All questions are compulsory.

There is no overall choice. However there is an internal choice has been provided in one question of two marks, one question of three marks and all three questions of 5 marks each.

You have to attempt only one of the choices in such questions.

(i) Questions 1 to 5 each one mark. **(VSA)**

(ii) Questions 6 to 10 each two mark. **(SA-I)**

(iii) Questions 11 to 22 each three mark. **(SA-II)**

(iv) Questions 23 is value based question and carry four marks. **(VBQ)**

(v) Questions 24 to 26 each five mark. **(LA)**

Q1. Write the dimension and SI unit of linear momentum. 1

Q2. What does the slope of velocity-time graph represent? 1

Q3. Give the magnitude and direction of net force acting on

(a) A drop of rain falling down with constant speed.

(b) A cork of mass 10 g floating on water. 1

Q4. Name the physical quantity which is expressed as 'the dot product of force and velocity'. Is it scalar or a vector quantity? 1

Q5. Is it necessary that the centre of mass of a body should always lie inside the body? Justify your answer. 1

Q6. In which of the following examples of motion can the body be considered approximately a point object:

(a) A railway carriage moving without jerks between two stations.

(b) A monkey sitting on the top of a man cycling smoothly on a circular track.

(c) A spinning cricket ball that turns sharply on hitting the ground, and

(d) Tumbling beaker that has slipped off the edge of a table? 2

Q7. (a) Define relative error. 2

(b) The error in the measurement of radius of sphere is

2%. What would be the error in the volume of the sphere? 2

Q8. State the law of conservation of linear momentum. Derive it from the Newton's second law of motion.

OR

Define coefficient of limiting friction.A body slides down an inclined plane having friction. Indicate the directions of frictional force and the reaction of the plane on the body.

2

Q9. A body of mass m is raised to a height h from the surface of the earth where the acceleration due to gravity on the surface of the earth. Prove that the loss in weight due to variation of g is approximately 2mgh/R. Where R is the radius of earth and $h<<R$ 2

Q10. The escape velocity on earth is 11.2 km/s. What will be its value on a planet having double the radius and eight times the mass of earth? 2

Q11. Explain why:

(a) It is easier to pull a lawn mower than to push it.

(b) A cricketer moves his hands backwards while holding a catch.

(c) The outer rail of a curved railway track is generally raised over the inner. 3

SESSION ENDING EXAMINATION – (2013-2014)

Physics (Set 1)

CBSE CLASS – XI

Time: 3 Hrs. M.M: 70

General Instructions:

(i) Question 1 to 5 one mark.

(ii) Question 6 to 10 each two mark.

(iii) Question 11 to 22 each three mark.

(iv) Question 23 is value based question and carry four marks.

(v) Questions 24 to 26 each five mark.

1. Write the dimension and SI unit of linear momentum.

2. What does the slope of velocity-time graph represent.

3. Give the magnitude and direction of net force acting on

(a) A drop of rain falling down with constant speed

(b) A cork of mass 10g floating on water.

4. Name the physical quantity which is expressed as 'the dot product of force and velocity'. Is it scalar or a vector quantity?

5. Is it necessary that the centre of mass of a body should always lie inside the body? Justify your answer.

6. In which of the following examples of motion can the body be considered approximately a point object:

(a) A railway carriage moving without jerks between two stations.

(b) A monkey sitting on the top of a man cycling smoothly on a circular track.

(c) A spinning cricket ball that turns sharply on hitting the ground, and

(d) Tumbling beaker that has slipped off the edge of a table?

7. (a) Define relative error.

(b) The error in the measurement of radius of sphere is 2%. What would be the error in the volume of the sphere?

8. State the law of conservation of linear momentum. Derive it from the Newton's second law of motion.

OR

Define coefficient of limiting friction. A body slides down an incline plane having friction. Indicate the directions of frictional force and the reaction of the plane on the body.

9. A body of mass m is raised to a height h from the surface of the earth where the acceleration due to gravity on the surface of the earth where the acceleration due to gravity on the surface of the earth. Prove that the loss in weight due to variation of g is approximately 2mgh/R. Where R is the radius of earth and H<<R.

10. The escape velocity on earth is 11.2 km/s. What will be its value on a planet having double the radius and eight times the mass of earth?

11. Explain why:

(a) It is easier to pull a lawn mower than to push it.

(b) A cricketer moves his hands backwards while holding a catch.

(c) The outer rail of a curved railway track is generally raised over the inner.

12. The displacement (in metre) of a particle moving along x-axis is given by x= 18t + $5t^2$.

Calculate:

(a) The instantaneous velocity at t = 2s.

(b) Average velocity between t = 2s and t = 3s.

(c) Instantaneous acceleration

13. Rain is falling vertically with a speed of 30 ms^{-1}. A woman rides a bicycle with a speed of 10 ms^{-1} in north to south direction. What is the relative velocity of rain with respect to the woman? What is the direction in which she should hold her umberalla to protect herself?

14. The velocity 'v' of water waves depends on the wavelength '¿', density of water 'p' and the acceleration due to gravity 'g'. Deduce by method of dimensions the relationship between these quantities.

15. A man weighs 70 kg. He stands on a weighing machine in a lift, which is moving

(a) Upwards with a uniform speed of 10m/s.

(b) Downwards with a uniform acceleration of 5 m/s^2.

(c) Upwards with a uniform acceleration of 5 m/s^2

What would be readings on the scale in each case?

16. Define angle of repose. Deduce its relation with coefficient of friction.

17. State and prove work-energy theorem.

18. Springs A and B are identical except that A is stiffer than B, i.e. constant $k_A > k_B$. In which spring is more work expended if they are stretched by the same amount.

19. (i) A child stands at the centre of a turn table with his two arms out stretched. The turn table is set rotating with an angular speed of 40 rpm. How much is the angular speed of the child change if he folds his hands back and thereby reduces his moment of inertia to $\frac{2}{3}$ times the initial value? Assume that the turntable rotates without friction.

(ii) Show that the child's new kinetic energy of rotation is more than the initial kinectic energy of rotation.

OR

State the law of conservation of angular momentum. Find the torque of a force $7i - 3j - 5k$ about the origin which acts on a particle whose position vector is $i + j - k$.

20. What is torque? Give its unit, show that it is equal to the product of force and the perpendicular distance of its line of action from the axis of rotation.

21. Discuss the variation of acceleration due to gravity with depth.

22. Define the term 'gravitational potential'. Derive an expression for the gravitation potential at a point in the gravitational field of the earth.

23. Raju saw his grandmother trying to clean a carpet. She was feeling difficulty in lifting the carpet. Raju helped his grandmother in cleaning the carpet by beating it with a stick.

SUBJECT - PHYSICS

CLASS - XI

Time : 3:00 Hrs. **M.M. 70 MARKS**

GENEREL INSTRUCTIONS:

1. All questions are compulsory.
2. There are 26 questions in all. Questions 1 to 5 carry 1 mark each, questions 6 to 10 carry 2 marks each, questions11 to 22 carry 3 marks each ,Q.No 23 carries 4 marks ,Q.Nos24 to 26 carry 5 marks each .
3. There is no overall choice. However, there is an internal choice in one question of 2 marks and one question of 3 marks and all three questions of five marks. you have to attempt only one of the given choices in such questions.
4. Use of calculator is not permitted.
5. Draw neat &labelled diagrams with pencil where –ever required.
6. You may use the following physical constants wherever necessary:

Boltzmann constant	$K_B = 1.38 \times 10^{-23}$ J/K
Acceleration due to gravity	$g = 10\ m/s^2$
Avogadro's Number	$N_A = 6.023 \times 10^{23}$/mole
Density of mercury	$\rho = 13.6 \times 10^3 kg / m^3$
Velocity of sound	$v = 340 m/s$

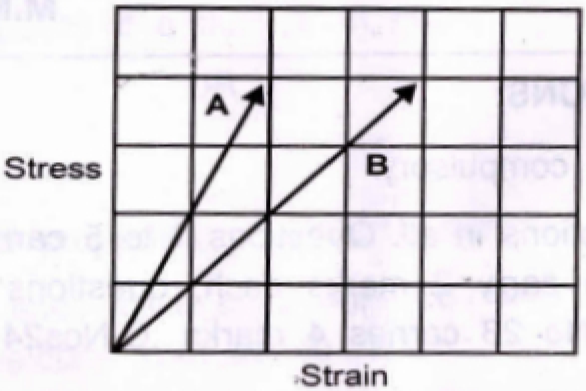
[P.T.O.]

1. Name the unit used to measure size of a nucleus of an atom. Also write its equivalence in SI system 1

2. What is the angle between two non- zero vectors if

$$|\vec{A}+\vec{B}| = |\vec{A}-\vec{B}|$$ 1

3. Stress - strain graph of two metals A & B are given below. Which one of the two is more elastic and why 1



4. "To keep a piece of paper horizontal in air, you should blow over, not under it." Explain. 1

5. A wave travelling along a string is given by

$$Y(x,t) = 0.005 \sin(80x - 3t)$$

Where the numerical values are in SI units. Symbols have their usual meanings Calculate the velocity of the wave. 1

6. Kinetic energy is given by the following formulae:

(i) K.E.=$3/16mv^2$

(ii) K.E.= $1/2\ mv^2 + ma$

Select the correct formula on the basis of dimensional arguments:2

7. Prove the following statement" for elevations which exceeds or fall short of 45° by equal angles, the ranges are equal".

[P.T.O.]

OR

A man of mass 70kg stands on a weighing scale in a lift which is:

(a) Moving down with the uniform acceleration of 5 m/s^2

(b) Moving upward with the uniform speed of 10 m/s

What would be the reading on the scale in each case 2

8. Name the Physical quantity equivalent to force in rotational motion. How is it related to force and give its unit? 2

9. State the no. of degrees of freedom possessed by a mono atomic molecule in space. Also give the expression for total energy possessed by it:-

(i) At given temperature, (ii)Calculate total energy of the atom at 300° K. 2

10. Draw temperature -volume graph for carbon dioxide gas near to room temperature

(a) As expected by Charles Law

(b) As observed experimentally 2

11. A ball is thrown vertically upwards with a velocity of 20m/s from the top of a building of height 40 meter from the ground.

(a) How high the ball reaches

(b) How long will it take to reach the ground

(c) The speed by which it strikes the ground 3

OR

A cricket ball is thrown at a speed of 28 m/s in a direction 30^0 above the horizontal Calculate :

PHYSICS

CLASS - XI

General Instructions

(a) Questions from question no. 1-4 carry 1 marks each, 5-12 carry 2 marks each, 13-27 carry 3 marks each and 28-30 carry 5 marks each.

(b) There is no overall choice but one choice is given in 2 marks question, two choice in 3 marks question and all three choices in five marks question

(c) You may use the following physical constant where ever necessary:

Speed of light $C = 3 \times 10^8\ ms^{-1}$

Gravitational constant $G = 6.6 \times 10^{-11}\ NM^2\ Kg^{-2}$

Gas constant $R = 8.314\ J\ Mol^{-1}\ k^{-1}$

Mass of electron $= 9.110 \times 10^{-31}$ Kg

Mechanical equivalent of heat $= 4.185\ J\ Cal^{-1}$

Standard atmospheric pressure $= 1.013 \times 10^5$ Pa

Absolute zero $0K = -273.15^0C$

Acceleration due to gravity $= 9.8\ Ms^{-2}$

Use of calculator is not permitted. However you may use log table, if required. Draw neat labelled diagram wherever necessary to explain your answer.

1. A light body and heavy body have equal momentum, which one have greater kinetic energy?
2. What does speedometer of a car indicates?
3. Write down the dimensions of viscosity coefficient
4. Why do we use ball-bearings?

5. How errors are combined in following mathematical operations of physical quantities?

   (i) Subtraction (ii) Product

6. Draw the Velocity - Time graph for following cases when (i) Object is moving in positive direction with acceleration (ii) An object is under free fall.

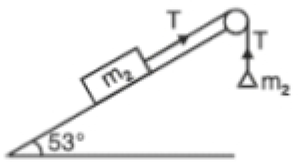
7. Derive the necessary relation for safest velocity of an automobile on a banked road radius r and friction coefficient μ.

8. If variation of position with time t is given by $x = a + bt + ct^2$. Write the dimensions of a, b & c.

9. The forces whose magnitude is in the ratio of 3:5 give a resultant of 35 N. If the angle b/w them is 60°. Find the magnitude of each force.

10. What is an impulse? A ball coming towards a batsman with a certain velocity U. He deflects the ball by an angle Q and its velocity increases to V. Draws A vector diagram to show initial momentum, find momentum and impulse.

11. In the given system of masses m_1 = 5 kg, and coefficient of friction for each constant is 0.2. Calculate the mass m_2, if m_1 is sliding down with an acceleration of 2 ms^{-2}. What will be the tension in the string?



12. The radius and length of a solid cylinder is measured as R = (10.0 +/- 02) cm, l = (20.0 + /- 0.5) cm. Calculate the volume and surface area of the cylinder and error in them.

13. A bomb is exploded into three fragments of mass 1:2:3. The fragment having lighter masses move with a speed of 40 m/s in mutually perpendicular to each other. Calculate the velocity of the third fragment.

14. If $\vec{A}\left(2\hat{i} + 2\hat{j} + 2\hat{k}\right)$ and $\vec{B} = \left(3\hat{i} + 4\hat{j}\right)$. Determine the vector having same magnitude as $\vec{B}$ and parallel to $\vec{A}$.

15. A force acting on an object is given by $\vec{F} = \left(3\hat{i} + 4\hat{j} - 6\hat{k}\right)$

N and the displacement made by it is given by $\vec{r} = \left(6\hat{i} - 2\hat{j} - \hat{k}\right)$. Calculate the work done and power if work is done in 2 s.

16. Define and prove conservation of linear momentum.

17. If the momentum of an object is increased by 50%, Calculate the percentage changes in its K.E.

OR

Two particles having mass ratio of 4:5 have same K.E. Calculate the ration of their linear momentum.

18. The velocity- Time relation of a particle is given by $V = (3t^2 - 2t - 1)$m/s. Calculate using calculus method, the position and acceleration of the particle when the velocity of the particle is zero. Given the initial position of the object is 5m.

19. Express 10J of energy in a new system of units in which 100g, 10 cm, 30 sec are the fundamental units. Determine which one of them is bigger unit of energy.

20. The escape velocity (v) of a body depends upon the mass (m) of body, gravitational acceleration (g) and radius (R) of the planet. Derive the relation for escape velocity dimensionally.

21. State and Prove Work- Energy Theorem. OR Define uniform velocity of an object moving along a straight line. What will be shape of velocity time and position-time graphs of such a motion?

22. If a composite physical quantity in terms of moment of inertia I, force F, velocity V, work W and length L is define as, $Q = (IF\ V^2/WL^3)$. Find the **dimension** of Q and identify it.

23. Explain why a man who fall from a height on a cemented floor receive more injury then when he fall from the same height on the heap of sand.

24. Is it possible to have collision in which all the kinetic energy is **lost**? If so cite an example.

OR

Subject : Physics - 2012
Theory

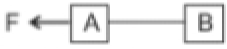
Time : 3 Hrs | Class : XI | M.M. : 70

General Instructions :

1. *All questions are compulsory.*
2. *There are 30 questions in total. Questions 1 to 8 carry one mark each , questions 9 to 18 carry two marks each, questions 19 to 27 carry three marks each and questions 28 to 30 carry five marks each.*
3. *There is no overall choice. However an internal choice has been provided in one question of 2 marks, one question of 3 marks and all of the three questions of 5 marks each. You have to attempt only one of the choices in such questions.*
4. *Use of calculator is not permitted.*

1. Write the dimensional formula for angular momentum.
2. Two bodies of masses m_1 and m_2 have equal momentum. Show which one has greater kinetic energy? Given $m_1 > m_2$.
3. State perpendicular axis theorem for a plane lamina.
4. State first law of themodynamics.
5. Two vessels of equal capacity have gases at the same temperature and pressure. The first vessel contains neon (monoatomic) the second contains chlorine (diatomic) gas. Which one has greater root mean squared speed.
6. Define coefficent of thermal conductivity.
7. Two sound sources produce 12 beats in 4 seconds. By how much do their frequencies differ?
8. What is resonance phenomenon?
9. The escape speed V of a body depends upon
   (i) acceleration due to gravity 'g' of the planet
   (ii) the radius of the planet R.
   Establish dimensionally the relationship in V, g and R.
10. A player throws a ball upwards with an initial speed *u* m/s.
    (i) What is the direction of acceleration during the upward motion of the ball?
    (ii) What is the direction of velocity during the upward motion?
    (iii) What are the velocity and acceleration of the ball at the highest point of its motion?
11. A body of mass 0.25 kg moving with velocity 12 m/s is stopped by applying a force of 0.6N. Calculate the time taken to stop the body.

Two bodies A and B of masses 10kg and 20kg respectively kept on a smooth, horizontal surface are tied to the ends of a light string. A horizontal force F = 600 N is applied to A. What is the tension on the string?



12. Two equal forces have their resultant equal to either. What is the inclination between them.
13. An elastic spring of spring constant k is compressed by an amount x. Obtain an expression for its potential energy.
14. State parallel axis theorem. What is the moment of incertia of a ring about an axis perpendicular to its plane and tangent to the ring.

    $M \rightarrow$ mass of ring, $r \rightarrow$ radius.
15. According to Newton's law of gravitation, the apple and the earth experience equal and opposite forces due to gravitation. But it is the apple that falls towards the earth and not vice versa. Why?
16. Draw energy distribution curve for a black body at two different temperatures T_1 and T_2 ($T_1 > T_2$). Write two conclusion that can be drawn from these curves.
17. Define mean free path of molecules in a gas. Obtain an expression for it.
18. Molar volume is the volume occupied by one mole of an ideal gas at standard temperature and pressure. Show that it is 22.4 litres. (S.T.P. : Pressure = 1 atmosphere, Temperature = 0°C)
19. A particle starts from origin at $t = 0$ with a velocity 5.0 $\hat{i}$ m/s and moves in xy plane under the action of a force which produces a constant acceleration of

    $(3.0\,\hat{i} + 2.0\,\hat{j})$ ms^{-2}

    (i) What is the y – coordinate of the particles and the instant its x – coordinate is 84m?

    (ii) What is the speed of the particle at this time?
20. Obtain an expression for maximum speed of a vehicle which can be achieved while taking a turn on a banked circular road.
21. Define the term "Coefficient of limiting friction between two surface." A body of mass 10kg is placed on an inclined surface at an angle of 30°. If the coefficient of limiting friction is $\frac{1}{\sqrt{3}}$, find the force required to just push the body up the inclined surface. The force is being applied parallel to the inclined surface.
22. Find the scalar and vector products of two vectors.

$$\vec{A} = 3\hat{i} - 4\hat{j} + 5\hat{k} \text{ and } \vec{B} = -2\hat{i} + \hat{j} - 3\hat{k}$$

23. Obtain an expression for kinetic energy of rotation of a body with angular velocity 'ω' and hence define moment of inertia.

24. Discuss the variation of acceleration due to gravity 'g' by deriving correct mathematical expression with depth from the surface of Earth.

    What is the value of 'g' at the centre of the Earth?

    OR

    State Kepler's laws of planetary motion.

    Let the speed of a planet at the peritetion be v_p and sun planet distance is r_p. Related (r_p, v_p) to the corresponding quantities at the aphelion (r_A, v_A).

25. Define surface energy of a liquid. How much work will be done in increasing the radius of a soap bubble from 1 cm to 5 cm (Surface tension = 73×10^2 N/m).

26. What is an adiabatic process? Find an expression for the work done in an adiabatic process.

27. If velocity of sound in gas is V at temperature T and pressure is P. How does it change if

    (i) temperature is doubled

    (ii) Pressure is halved

    (iii) Humidity in gas increases

28. Define circular motion. Derive an expression for velocity and acceleration (centripetal) for a body executing uniform circular motion.

    OR

    What is a projectile? Show that the path of a projectile is parabolic. Obtain an expression for

    (i) maximum height

    (ii) Time of flight

    (iii) Horizontal range when the projectile is fired at an angle θ with the horizontal.

29. State and prove Bernoulli's Theorem. Explain the following :

    (i) Can Bernoulli's equation be used to describe the flow of water through a rapid river?

    (ii) Does it matter if one uses gauge pressure instead of absolute pressure by applying Bernoulli's equation.

    OR

    (i) What is the phenomenon of capillarity? Derive an expression for the size of liquid in a capillary tube.

    (ii) What will happen if the length of the capillary tube is smaller then the height to which the liquid rizes? Explain briefly.

EXAMINATION PAPER : 2011

Time : 3 hours **Maximum Marks : 70**

General Instructions :

(i) All questions are compulsory.

(ii) There are 30 questions in total. Questions 1 to 8 carry one mark each, questions 9 to 18 carry two marks each, questions 19 to 27 carry three marks each and question 28 to 30 carry five marks each.

(iii) There is no overall choice. However, an internal choice has been provided in the one question of two marks; one question of three marks and all three questions of five marks each. You have to attempt only one of the choices in such questions.

(iv) Use of calculators is not permitted.

(v) Please write down the serial number of question before attempting it.

(vi) You may use the following values of physical constants wherever necessary.

Boltzmann's constant $K = 1.38 \times 10^{-23}\ JK^{-1}$

Avogadro's number $N_A = 6.022 \times 10^{23}/mol$

Radius of Earth $R_e = 6400$ km.

$= 1.013 \times 10^5$ Pa

1 Atmospheric Pressure $g = 9.8\ m/s^2$

$R = 8.3\ J\ mol^{-1}\ K^{-1}$

1. Write the dimensional formula of torque. 1

2. Draw velocity-time graph for an object, starting from rest. Acceleration is constant and remains positive. 1

3. Arrange increasing order the tension T_1, T_2 and T_3 in the figure.

4. Why there is lack of atmosphere on the surface of moon? 1

5. The triple point of carbon dioxide is 216.55 K. Express this temperature on Fahrenheit scale. 1

6. In an open organ pipe, third harmonic is 450 Hz. What is the frequency of fifth harmonic?

7. Which type of substances are called elastomers? Give one example 1

8. A simple harmonic motion is described by $a = -16\,x$ where a is acceleration, $x \rightarrow$ displacement in m. What is the time period? 1

9. Percentage error in the measurement of height and radius of cylinder are x and y respectively. Find percentage error in the measurement of volume. Which of the two measurements height or radius need more attention? 2

OR

The length and breadth of a rectangle are measured as $(a \pm \Delta a)$and$(b \pm \Delta b)$ respectively. Find (i) relative error. (ii) absolute error in the measurement of area. 2

10. An object moving on a straight line covers first half of the distance at speed v and second half of the distance at speed 2v. Find (i) average speed, (ii) mean speed.

11. An object moving on a circular path in horizontal plane. Radius of the paths is *r* and constant speed is v. Deduce expression for centripetal acceleration. 2

12. Find the height from the surface of earth at which weight of a body of mass *m* will be reduced by 36% of its weight on the surface. (R_e = 6400 km) 2

13. Define gravitational potential. Give its S.I. unit.

14. An engine has been designed to work between source and sink at temperature 177°C and 27°C respectively. If energy input is 3600 J. What is the work done by the engine? 2

15. Explain :

    (i) Why does the air pressure in a car tyre during driving increase?

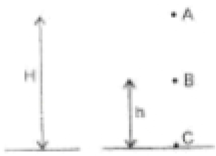
    (ii) Why coolant used in a chemical plant should have high specific heat? 2

16. Calculate the work done in blowing a soap bubble from a radius of 2 cm to 3 cm. The surface tension of the soap solution is 30 dynes cm^{-1}. 2

17. Show that Newton's second law of motion is the real law of motion. 2

18. A block initially at rest breaks into two parts of masses in the ratio 2 : 3. The velocity of smaller part is $\left(8\hat{i} + 6\hat{j}\right)$ m/s. Find the velocity of bigger part. 2

19. A body of mass *m* is released in vacuum from the position A at a height H above the ground. Prove that sum of kinetic and potential energies at A, B and C remains constant. 3



20. Give two points of difference between elastic and inelastic collisions. Two balls A and B with A in motion initially and B at rest. Find their velocities after collision (perfectly elastic). Each ball is of mass "m". 3

21. A liquid is in streamlined flow through a tube of non-uniform cross-section. Prove that sum of its kinetic energy, pressure energy and potential energy per unit volume remains constant. 3

22. Give reason :

(i) fog particles appear suspended in atmosphere.

(ii) two boats being moved parallel to each other attract.

(iii) bridges are declared unsafe after long use. 3

23. State Kepler's law of Planetary motion. Name the physical quantities which remain constant during the planetary motion. 3

24. What is the law of equipartition of energy? Determine the value of γ for diatomic gas N_2 at moderate temperature. 3

25. Show that for small oscillations the motion of a simple pendulum is simple harmonic. Drive an expression for its time period. Does it depend on the mass of the bob? 3

EXAMINATION PAPER : 2010

Time : 3 hours **Maximum Marks : 70**

General Instructions :

(i) All questions are compulsory.

(ii) There are 30 questions in total. Questions 1 to 8 carry one mark each, questions 9 to 18 carry two marks each, questions 19 to 27 carry three marks each and question 28 to 30 carry five marks each.

(iii) There is no overall choice. However, an internal choice has been provided in the one question of two marks; one question of three marks and all three questions of five marks each. You have to attempt only one of the choices in such questions.

(iv) Use of calculators in not permitted.

(v) Please write down the serial number of question before attempting it.

(vi) You may use the following values of physical constant wherever necessary.

Boltzmann's constant $K = 1.38 \times 10^{-23}\ JK^{-1}$

Avogadro's number $N_A = 6.022 \times 10^{23}$/mol

Radius of Earth R = 6400 km.

1. Express one micron in metre. 1

2. What does the slope of velocity-time graph represent? 1

3. Are the magnitude and direction of $\left(\vec{A} - \vec{B}\right)$ same as that of $\left(\vec{B} - \vec{A}\right)$? 1

4. What is the principle of working of a rocket? 1

5. Why do we slip on a rainy day? 1

6. What is the source of the kinetic energy of the falling rain drop? 1

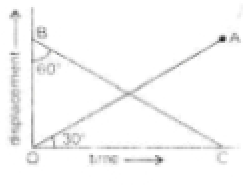
7. Two bodies move in two concentric circular paths of radii r_1 and r_2 with same time period. What is the ratio of their angular velocities? 1

8. State second law of thermodynamics. 1

9. If $x = at + bt^2$, where x is in metre and t in hour, what will be the unit of 'a' and 'b'? 2

10. The displacement-time graph of two bodies P and Q are represented by OA and BC respectively. What is the ratio of velocities of P and Q?

$\angle OBC = 60°$ and $\angle AOC = 30°$



OR

A car moving with a speed of 50 kmh^{-1} can be stopped by brakes after at least 6 m. What will be the minimum stopping distance, if the same car is moving at speed of 100 kmh^{-1} ? 2

11. A particle of mass m is moving in an horizontal circle of radius 'r', under a centripetal force equal to (k/r^2), where k is a constant. What is its potential energy? 2

12. Two solid spheres of the same are made of metals of different densities, which of them has larger moment of inertia about its diameter? Why? 2

13. If suddenly the gravitational force of attraction between the earth and a satellite revolving around it becomes zero, what will happen to the satellite? 2

14. When air is blown in between two balls suspended close to each other, they are attracted towards each other. Give reason. 2

15. What is an isothermal process?

    Also give essential conditions for an isothermal process to take place. 2

16. Calculate the fall in temperature of helium initially at 15°C, when it is suddenly expanded to 8 times its volume. Given $\gamma = 5/3$. 2

17. Three vessels of equal capacity have gases at the same temperature and pressure. The first vessel contains neon (monoatomic), the second vessel contains chlorine(diatomic) and third contain polyatomic gas. Do the vessel contain equal number of molecules? Is the root-mean square speed of molecules same in three cases? 2

18. A particle is in linear simple harmonic motion between two points A and B, 10 cm apart, Take the direction from A to B as positive direction and give the signs of velocity and acceleration on the particle when it is

(i) at the end B

(ii) at 3 cm away from A going towards B 2

19. Draw displacement-time, velocity-time and acceleration-time graphs for a particle executing simple harmonic motion.

OR

A bat emits ultrasonic sound of frequency 1000 kHz in air. If the sound meets a water surface, what is the wavelength of (a) the reflected sound (b) transmitted sound? Speed of sound in air $v_a = 340\ ms^{-1}$, in water $v_w = 1486\ ms^{-1}$ 3

20. (i) Define Absolute Zero.

(ii) Deduce the dimensional formula for R, using ideal gas equation PV = nRT

(iii) Find degree of freedom of a monoatomic gas. 1+1+1=3

21. What is the temperature of the steel-copper junction in the steady state of the system shown in figure. The area of cross-section of steel rod is twice that of the copper rod, $K_{steel} = 50.2\ Js^{-1}\ m^{-1}K^{-1}$, $K_{cu} = 385\ Js^{-1}\ m^{-1}K^{-1}$

3

22. Define the term gravitational potential. Give its S.I. unit. Also derive expression for the gravitational potential energy at a point in the gravitational field of the earth. 3

23. Write S.I. unit of torque and angular momentum. Also deduce the relation between angular momentum and torque. 1/2+1/2+2=3

24. Show that the total mechanical energy of a freely falling body remains constant through out the fall. 3

25. Two masses 8 kg and 12 kg are connected at the two ends of a light inextensible string that goes over a frictionless pulley. Find the acceleration